

Printed Payload Module — Specification

The 3D-printed "top" that carries the marker, cameras, lidar, Pi, and trigger, and bolts to the gimbal's **tilt-output interface plate (PA-4)**. The gimbal is payload-agnostic and built separately; this module is the only scan-driven part of the system.

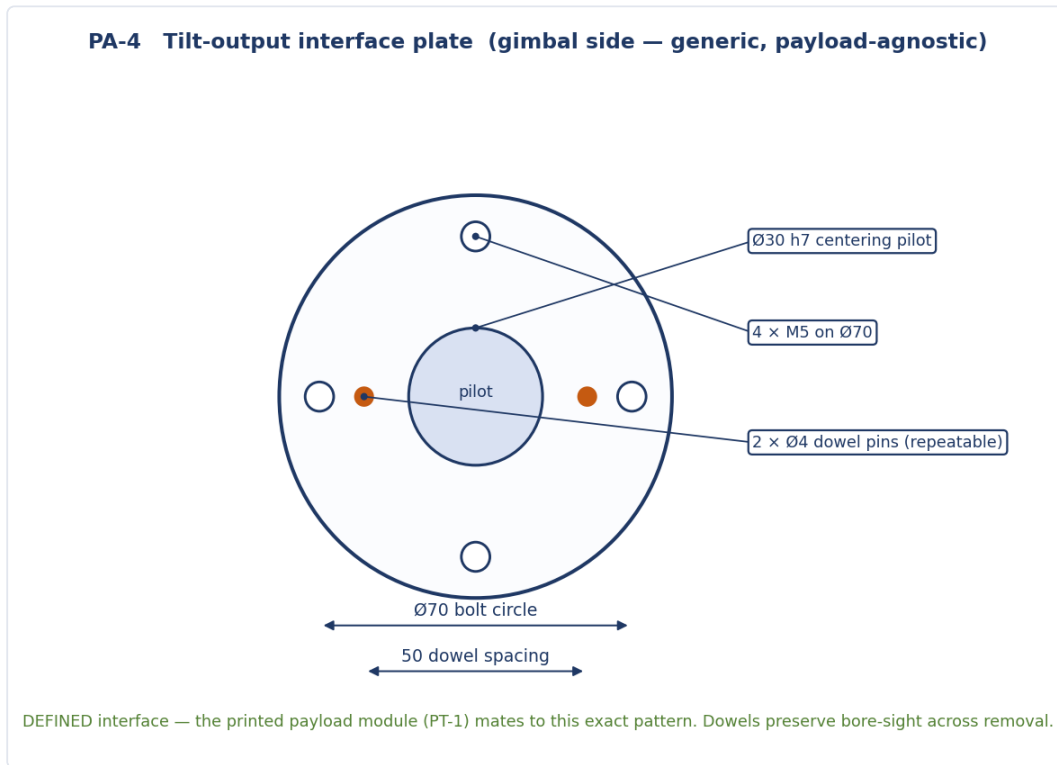
Companion files: `Gimbal_Assembly_Guide.pdf` (the gimbal it mounts to), `Turret_Full_Parts_List.xlsx`.

1. Mounting interface (mates to PA-4)

The module's underside carries the mating pattern of the gimbal interface — **fixed, not adjustable**:

- Bolt circle **Ø70 mm**, 4 × M5 clearance
- Centering pilot **Ø30 mm** (recess over the gimbal's h7 spigot)
- **2 × Ø4 mm dowel holes** at ±25 mm

The dowels locate the module repeatably, so it can be removed and reinstalled without losing bore-sight calibration. Use **heat-set inserts** or metal fasteners through the printed part into the interface — never thread a metal bolt directly into plastic.

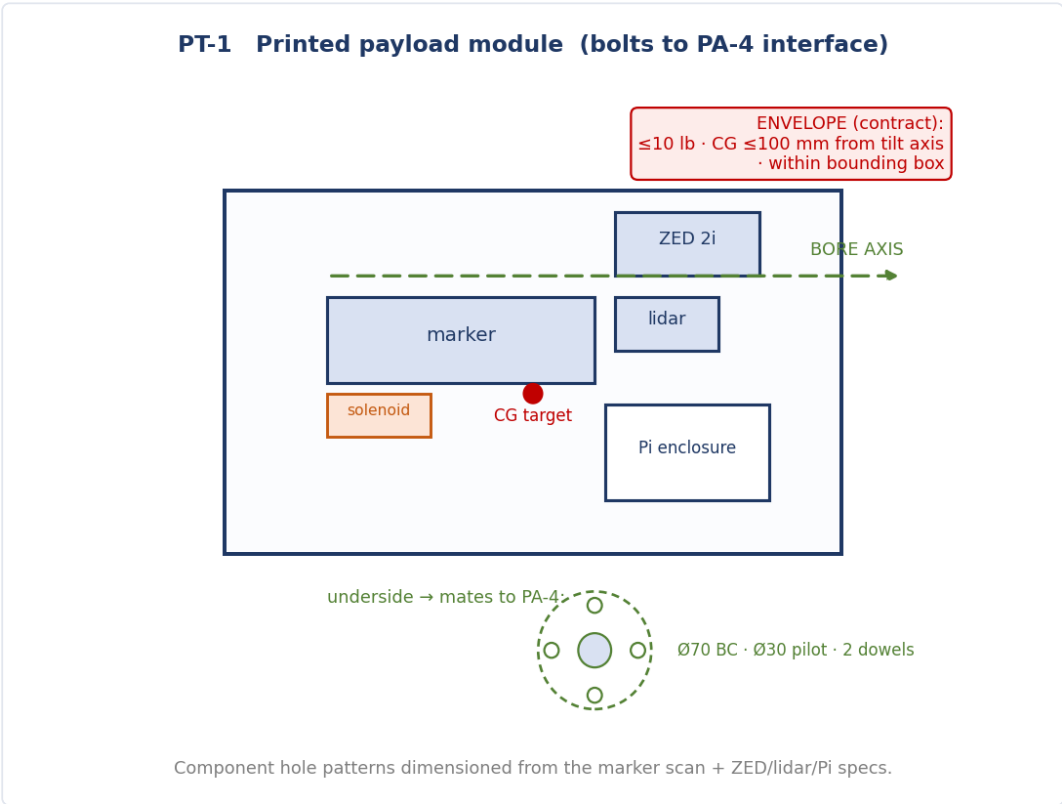


2. Envelope — the contract with the gimbal

The module must stay inside the envelope the gimbal was sized for. If it grows past this, the gimbal must be resized.

Constraint	Limit
Total mass (everything on the module)	$\leq 10 \text{ lb}$
Combined CG distance from tilt axis	$\leq 100 \text{ mm}$
Bounding box (tilt clearance)	within frame envelope (confirm vs. final gimbal frame)

3. Component layout



- **Marker** on its cradle, with the **solenoid/trigger bracket** referencing the actual trigger (from scan).
- **ZED 2i** and **TF03 lidar** co-aligned with the muzzle on **one bore axis**.
- **Pi enclosure** (HAT stack + active cooler) positioned to help bring the combined CG over the tilt axis.
- Heavy items placed to land the **CG within 100 mm of the tilt axis**.

4. Component mounting (sources)

Component	Interface	Source
Marker cradle + trigger bracket	clamp + solenoid mount	Marker scan (trigger, mounting surface, muzzle)
ZED 2i	1/4-20 + flat	datasheet + caliper check
TF03 lidar	M3 pattern	datasheet + caliper check
BNO085 IMU	M2.5 standoffs	breakout dims
Pi 5 + HAT + cooler	M2.5 standoffs + port cutouts	measured envelope

5. Bore-sight

Mount the ZED, lidar, and marker muzzle to a common bore axis so the perception range and the projectile path agree. The hand-eye (checkerboard) calibration refines the residual; the dowel-located interface preserves it across module removal.

6. Print & material notes

- **Material:** CF-nylon or ASA for stiffness and outdoor durability; PETG acceptable for lighter loads. The module is semi-structural — it resists marker recoil — so design with ribs and generous walls/infill, not a thin plate.
 - **Metal threads:** heat-set inserts for the 1/4-20 (ZED), M3 (lidar), M2.5 (Pi/IMU), and the M5 interface bolts.
 - **Marker recoil:** if recoil is significant, back the marker cradle with a metal sub-bracket rather than relying on plastic alone.
 - **Cable management:** service loop at the tilt axis; route signal away from motor/solenoid power.
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7. Inputs needed to finalize

1. **Marker scan** — trigger + guard, mounting surface, muzzle/bore line, full envelope (include hopper/air **if** they ride on the module).
2. ZED, lidar, Pi-stack, IMU dimensions (caliper/datasheet).
3. Confirmation of the bounding-box limit once the gimbal frame is sized.

With those, this becomes a printable **STL/STEP** — closing the loop to the original goal.